



PROGRAMMING WITH PYTHON

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Computer Applications

PROGRAMMING WITH PYTHON

Exception Handling, File I/O, Modules and Packages

File Handling – Text

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File Handling

- Python supports the file-handling process.
- The input was taken from the console and written back to the console to interact with the user.
- Users can easily handle the files, like read and write the files in Python.
- File handling is also effortless and short in Python.



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File Handling – Text Files

- Open the file using Python's built-in `open()` function.
- This function creates a file object, which would be utilized to call other support methods associated with it.
 - `file object = open(file_name [, access_mode][, buffering])`



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File Handling – Text Files

- `file_name`
 - The `file_name` argument is a string value that contains the name of the file that is accessed.



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File Handling – Text Files

- `access_mode`
 - The `access_mode` determines the mode in which the file has to be opened, i.e., read, write, append, etc.
 - This is optional parameter and the default file access mode is read (r).



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File Handling – Text Files

	Description
r	Opens a file for reading only. The file pointer is placed at the beginning of the file. This is the default mode.
r+	Opens a file for both reading and writing. The file pointer placed at the beginning of the file.
w	Opens a file for writing only. Overwrites the file if the file exists. If the file does not exist, creates a new file for writing.



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File Handling – Text Files

	Description
w+	Opens a file for both writing and reading. Overwrites the existing file if the file exists. If the file does not exist, creates a new file for reading and writing.
a	Opens a file for appending. The file pointer is at the end of the file if the file exists. If the file does not exist, it creates a new file for writing.



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File Handling – Text Files

	Description
a+	Opens a file for both appending and reading. The file pointer is at the end of the file if the file exists. If the file does not exist, it creates a new file for reading and writing.



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File Handling – Text Files

- buffering
 - If the buffering value is set to 0, no buffering takes place.
 - If the buffering value is 1, line buffering is performed while accessing a file.
 - If the buffering value is specified as an integer greater than 1, then buffering action is performed with the indicated buffer size.
 - If negative, the buffer size is the system default(default behavior).



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File Handling – Text Files

- Read a file
 - `file.read(size)`
 - Size
 - An integer that specifies the number of bytes to read from the file.
 - If omitted or set to -1, it reads the entire file.



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File Handling – Text Files

- Read a file
 - `file.read(size)`
 - Returns a string containing the data read from the file
 - Returns an empty string if read beyond the end of file



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File Handling – Text Files

- Read a line
 - `file.readline(size=-1)`
 - size
 - An integer that specifies the maximum number of characters (or bytes in binary mode) to read from the line.



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File Handling – Text Files

- Read a line
 - `file.readline(size=-1)`
 - size
 - If size is omitted or set to -1, it reads the entire line, up to the newline character (`\n`).
 - If the specified size is smaller than the length of the line, only size characters will be returned.



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File Handling – Text Files

- Read a line
 - Output is a string representing the line read from the file.
 - Includes the newline character (`\n`) at the end of the line unless the end of the file is reached or the line is shorter than size.



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File Handling – Text Files

- Read line by line
 - `file.readlines(hint=-1)`
 - `hint`
 - An integer specifying the maximum number of bytes to read.
 - If omitted or set to -1, it reads all lines from the file.
 - Useful for limiting the amount of data read into memory.



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File Handling – Text Files

- Read line by line
 - `file.readlines(hint=-1)`
 - It returns a list of strings, where each string represents a line in the file (including the newline character `\n`).



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File Handling – Text Files

Function	Reads	Returns	Usecase
read()	Entire file or specified characters	Single string	Use when whole content is needed as one block.
readline()	One line at a time	Single string (one line)	Use when processing one line at a time.
readlines()	All lines at once	List of strings (each line)	Use when all lines are needed as separate items.



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File Handling – Text Files

- Write text
 - `file.write(string)`
 - The string to be written to the file.
 - It returns the number of characters written.



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File Handling – Text Files

- Write a text
 - `file.writelines(iterable)`
 - An iterable (e.g., list or tuple) of strings to be written to the file.
 - Each string in the iterable is written as-is without adding a newline unless explicitly included.



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File Handling – Text Files

Feature	write()	writelines()
Purpose	Writes a single string to the file.	Writes multiple strings from an iterable to the file.
Input	A single string.	An iterable of strings (e.g., list, tuple).
Adds Newline	No, unless included in the string.	No, unless included in each string in the iterable.



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File Handling – Text Files

- **Open a file**

```
f=open('file1.txt','r')
```

```
print('Name of the file is ', f.name)
```

Name of the file is file1.txt

```
print(f'This file is opened in {f.mode} mode')
```

This file is opened in r mode



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File Handling – Text Files

Read a file

```
: str=f.read()
```

```
: print(str)
```

This is a file that is used for reading.

The file contains all information that needs to be read.

It contains lines of information to be read.

The file has to have a name, size and other details.

```
: f.close()
```



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File Handling – Text Files

```
f.close()
```

```
f=open('file1.txt','r')
```

Readline function

```
str1=f.readline(10)
```

```
print(str1)
```

This is a

```
f.close()
```




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File Handling – Text Files

Readlines Function ¶

```
f=open('file1.txt','r')
```

```
str1=f.readlines(10)
```

```
print(str1)
```

```
['This is a file that is used for reading.\n']
```

```
f.close()
```



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File Handling – Text Files

Write Function

```
with open('file2.txt', 'w') as file:  
    characters_written = file.write("Hello, World!")  
    print(f"The Text written \n: {characters_written}")
```

The Text written
: 13

Writelines Function

```
lines = ["First line\n", "Second line\n", "Third line\n"]  
with open('file3.txt', 'w') as file:  
    file.writelines(lines)
```

```
file=open('file3.txt','r')
```

```
s=file.read()
```

```
print(s)
```

First line
Second line
Third line



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File Handling – Text Files

```
print("File is closed or not ?", file.closed)
```

File is closed or not ? False

```
file.close()
```

```
print("File is closed or not ?", file.closed)
```

File is closed or not ? True



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Recap

- File handling
 - Text Files



THANK YOU

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